Anchal Thool

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Education

New York University, Tandon School of Engineering — **New York, NY**
M.S. in Management of Technology (STEM) — Sep 2024 – May 2026

Veermata Jijabai Technological Institute (VJTI) — **Mumbai, India**
B.Tech in Electronics & Telecommunication Engineering — Aug 2018 – May 2022

Technical Skills

Languages & Databases: Python, Java, Go, C, C++, TypeScript, JavaScript, SQL, PostgreSQL, MongoDB, DynamoDB, Redis
AI / Machine Learning: LLM Applications, RAG, Gemini, Structured Outputs, Prompt Engineering, Machine Learning, NLP
Backend & Distributed Systems: REST APIs, gRPC, Microservices, Kafka, Event-Driven Architecture, OAuth 2.0, React, Next.js
Cloud & DevOps: AWS (Lambda, EventBridge, SQS, IAM), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions
Observability & Foundations: Prometheus, Grafana, CloudWatch, SonarQube, Data Structures, Algorithms, OOP, System Design

Professional Experience

Software Engineer — **Jul 2022 – Aug 2024**
Telstra — Pune, India

- **Account Deprovisioning Automation:** Architected a fault-tolerant, event-driven AWS workflow using Lambda, EventBridge, SQS, retry handling, audit reporting, CloudWatch, and CloudFormation, automating cleanup of 13K+ inactive accounts and cutting manual administration effort by 87%.
- **Access Provisioning Workflow:** Designed a Kafka-based integration across Jira, Slack, and AWS IAM to automate approval-driven onboarding and permission updates in a high-throughput, event-driven environment.
- **Credential Leak Monitoring:** Built automated Confluence credential-leak monitoring with Python and CloudWatch, detecting exposed tokens, triggering remediation workflows, and verifying cleanup through repeat scans to reduce security exposure risk.
- Supported Jira/Confluence upgrades using Jenkins, Ansible, & AMI-based deployments, improving release consistency and reducing manual effort.
- **Backend Quality & Testing:** Enforced 95%+ backend unit-test coverage with Bamboo CI and SonarQube and led code reviews on interface design and testing strategy, cutting production defects by 35%.

Backend Engineer Intern — **Jan 2022 – Jun 2022**
Agarkar Media — Nagpur, India

- **API & Authentication Platform:** - Built an authentication backend with OAuth 2.0/OIDC and PKCE, implementing short-lived JWT access tokens, rotating refresh tokens, HttpOnly cookies, OTP-based MFA, token revocation, and rate-limited auth endpoints for secure session management and protected API access.

Full Stack Engineer Intern — **May 2021 – Sep 2021**
Sansoft Technologies — Nagpur, India

- **Event-Driven Notification Pipeline:** Designed an email notification architecture using Kafka for asynchronous event delivery, a dedicated worker service for retries and idempotent processing, and SendGrid for transactional email delivery, isolating notification failures from core application workflows.

Projects

Tabby – AI Receipt-Splitting Platform | Live Demo — Next.js, React, TypeScript, Gemini, Vercel

- **Multimodal Document Intelligence:** Built a Gemini-powered receipt understanding pipeline with image parsing, structured generation, and schema validation to extract reliable item-level transaction data from multiple receipt images.
- **LLM Reliability & Post-Processing:** Designed deterministic validation and post-processing for duplicate items, shared purchases, taxes, fees, discounts, and payer attribution, converting probabilistic model outputs into consistent downstream calculations.

CloudForge AI | GitHub — Python, FastAPI, AWS, Terraform, RAG, LangChain

- **Natural Language to Infrastructure:** Built an LLM agent that converts English requests into Terraform/CloudFormation, grounding Gemini with RAG over 21 security policies and a self-repair loop for validation and 22 Checkov policy failures.
- **Safe Deployment Workflow:** Engineered a human-in-the-loop pipeline that applies only the reviewed plan, gates IAM changes, and blocks prompt-injection attacks; validated with 96 tests and a 24-case adversarial eval harness.

RentAnything – Peer-to-Peer Rental Marketplace | GitHub — Next.js, Node.js, PostgreSQL, Redis, AWS S3

- **Marketplace Backend APIs:** Designed and built 15+ REST APIs for listings, bookings, favorites, reviews, availability, and owner workflows with backend validation, authorization, and controlled state transitions.
- **Concurrency-Safe Booking:** Prevented double-booking under simultaneous requests using transactional locking and availability re-checks, validating correctness with concurrent k6 load tests.